GridSense: Complete Knowledge Base

Team: Quantum Minds | Institution: Jain Institute of Technology | Challenge No. 12

PROJECT IDENTITY

- **Project Name**: GridSense
- **Tagline**: Campus Intelligent Energy Optimization Platform
- **Team**: Quantum Minds
- **Institution**: Jain Institute of Technology (JIT)
- **Challenge**: No. 12 / The Campus That Wastes Energy After Hours
- **Event**: Hackathon O1

SECTION 1: EXPLAIN THE CHALLENGE

The Situation

Walk through any Indian engineering campus at 10:00 PM. The labs are empty, but the lights are blazing. Projectors hum in vacant lecture halls. In one department, five air conditioners run at full blast, cooling an empty seminar room. The library keeps three full floors lit and chilled for a handful of students clustered in one corner.

The administration knows electricity bills are through the roof. But they do not know where the waste is happening, why it is happening, or what to do about it without disrupting academic life.

Key Statistics

- 40% of total campus electricity is consumed after 6:00 PM with minimal occupancy
- Average monthly electricity bill for a mid-sized engineering college: INR 8 Lakhs or more
- 30% to 45% of total consumption is eliminable with zero disruption using intelligent automation

The Core Conflict

Campus energy systems run on static timers and manual switches. Facility managers cannot simply cut power without risking safety or disrupting research. Manual guard patrols are inconsistent and happen hours too late. Administrators receive bulk monthly utility bills with no actionable per-room or per-hour breakdown.

The Fundamental Mismatch

Energy infrastructure is static and scheduled. Human presence is dynamic and unpredictable. GridSense bridges this gap by making every energy decision data-driven, occupancy-aware, and contextually intelligent.

Campus Layout (Typical Indian B.Tech College)

- Block A: Administrative offices, faculty cabins, executive seminar halls, auditorium
- Block B: First and second-year lecture halls, large central drawing and seminar halls
- Block C: Specialized computer and hardware labs, departmental classrooms (CSE, ECE, Mech)
- Hostel Block: High-density living, common bathrooms with geysers, study rooms, dining hall
- Connecting Infrastructure: Wide arterial hallways (50 to 100 metres long), parking bays, outdoor perimeter lighting, central elevator banks

SECTION 2: WHAT IS THE PROBLEM

The Decoupling of Energy and Occupancy

In a standard Indian B.Tech campus, the physical layout creates four distinct energy waste zones.

Zone A: Academic Blocks (The Empty-Room Full-Load Problem) Block A has one class at 7:00 PM while Block C has five. The entire Block A corridor lighting system (80 metres), all HVAC units, and elevator standby draws remain active for a single class. Simultaneously, lecture halls run air conditioning at 18 to 20 degrees C with 10 students for a class scheduled for 60. Root Cause: Static allocation, no consolidation intelligence.

Zone B: Computer Labs (The Phantom PC Drain) A 60-PC computer lab left in idle desktop mode draws between 80 and 120 Watts per machine. After 6:00 PM, with no active users, all 60 machines and their UPS inverters continue drawing power. The dedicated split AC runs to cool the heat these idle machines generate, compounding the waste. Root Cause: No idle detection or scheduled shutdown policy.

Zone C: Hostels (Morning kVA Spike and 24-Hour Utility Losses) Between 6:00 AM and 8:30 AM, all hostel wing geysers (2 to 3 kW each) fire simultaneously, pushing the campus above its contracted Maximum Demand (kVA) threshold. The state electricity board charges heavy penalty tariffs for every kVA above the MD contract threshold. Meanwhile, 15 commercial water coolers across all blocks run 24-hour active refrigeration, chilling water at 3:00 AM in locked, empty buildings. Root Cause: Static timers, no demand management, no occupancy interlocking.

Zone D: Infrastructure (Elevators and WiFi Running 24/7) All 4 elevators remain active with full cabin lighting, ventilation fans, and motor drives on standby through the night. 150 enterprise WiFi access points draw 15 to 30 Watts each via PoE, running their full 5 GHz radios constantly even during the 11:00 PM to 6:00 AM window when academic blocks are locked. Root Cause: No time-aware or occupancy-aware infrastructure scheduling.

SECTION 3: GAPS AND CHALLENGES

Gap 1: The Raw Data Trap

Facility managers receive end-of-month bulk utility bills or unmanageable 15-minute interval smart meter spreadsheets. They are told the campus consumed 48,000 kWh in October. This information is completely useless for action. Knowing kWh spiked is useless without knowing which room, which floor, and whether any person was present. What is needed: "Block B Seminar Hall 2 wasted 42 kWh on Wednesday evening with zero registered occupants" is 1,000 times more useful than a raw kWh total.

Gap 2: The Hardware Cost Barrier

Every existing smart campus proposal begins with a full IoT sensor retrofit: PIR motion sensors in each classroom, occupancy sensors per HVAC unit, door contact sensors per lab. For a campus with 200 rooms, 60 lab PCs, 4 elevator shafts, and 150 APs, the hardware cost runs into tens of lakhs with an 18 to 24 month deployment timeline. What is needed: A solution that extracts maximum intelligence from infrastructure the campus already owns: its enterprise WiFi controller, its ERP timetable database, and its main smart meter.

Gap 3: The All-or-Nothing Dilemma

Existing campus energy policies are binary: either power stays on all night, or someone manually switches it off. Abrupt automated shutoffs anger professors and disrupt students. Manual policies have a documented compliance rate below 30% in Indian college settings. What is needed: A graduated, 5-tier intervention system that starts with gentle notifications, escalates through automated setpoint adjustments, and only reaches full isolation as a last resort on definitively empty zones.

Gap 4: Spatial Fragmentation is Invisible

No current system understands that having 1 class in Block A and 5 in Block C means Block A should be shut down and that lone class relocated. Spatial fragmentation is the single largest driver of after-hours energy waste in Indian campuses, yet it is entirely invisible to conventional Building Management Systems. What is needed: A spatial consolidation engine that thinks at the building, wing, floor, and corridor zone level simultaneously and can re-route academic schedules to cluster activity and isolate empty infrastructure.

SECTION 4: SOLUTION AND ARCHITECTURE

GridSense Core Principle

GridSense is a weighted, multi-input scoring engine that continuously monitors every energy-consuming zone across a campus. It assigns a real-time Waste Index Score (WIS) of 0 to 100 to each zone every 15 minutes. The higher the score, the more urgent and aggressive the automated intervention.

The system does not treat all inputs as equal. Every data signal carries a pre-calibrated weight. The weighted combination produces the WIS.

THE GRIDSENSE SCORING ENGINE

Input Categories and Weights

Occupancy Signals (Sub-Score Weight: 0.60) Highest weight because occupancy is the single strongest predictor of wasted energy.

- WiFi Connected Clients per AP zone (Weight: 0.25) | Source: Campus Network Controller SNMP
- ERP Timetable Status (Weight: 0.20) | Source: College ERP / Timetable API
- RFID / Biometric Entry Log (Weight: 0.10) | Source: Door access controller
- Camera Occupancy Count (Weight: 0.05) | Source: IP Camera with CV inference (optional)

Climate Signals (Sub-Score Weight: 0.23) Determines how much energy is appropriate for a given occupancy, not whether it should run at all.

- Outdoor Ambient Temperature (Weight: 0.08) | Source: Weather API (OpenMeteo / IMD)
- Outdoor Lux Level / Daylight (Weight: 0.07) | Source: Solar irradiance API or lux sensor
- Indoor Temperature (Weight: 0.05) | Source: Smart thermostat / BMS sensor

- Humidity Level (Weight: 0.03) | Source: IoT sensor or weather API

Energy Draw Signals (Sub-Score Weight: 0.21) Confirms whether the zone is actually overconsumning relative to its occupancy-adjusted expected baseline.

- Real-Time Zone Power Draw in kW (Weight: 0.08) | Source: Smart sub-meter on Distribution Board
- Deviation from Expected Baseline (Weight: 0.06) | Source: GridSense internal prediction model
- Peak Demand Proximity as % of MD limit (Weight: 0.04) | Source: Smart meter demand register
- Tariff Period Flag (Weight: 0.03) | Source: Time-of-Day tariff calendar

Schedule and Context Signals (Sub-Score Weight: 0.12) Provides temporal context and prevents false positives during officially scheduled but sparsely attended events.

- Time of Day (Weight: 0.05) | Source: System clock
- Day Type: holiday vs weekday (Weight: 0.04) | Source: Academic calendar API
- Event / Extension Override Active (Weight: 0.03) | Source: GridSense portal override flag

Master Waste Index Score Formula

$\text{WIS} = (\text{Occupancy\_Score} \times 0.60) + (\text{Climate\_Score} \times 0.23) + (\text{Energy\_Score} \times 0.21) + (\text{Schedule\_Score} \times 0.12)$ Each sub-score is normalized on a 0 to 10 scale where 0 = ideal and 10 = maximum waste. Final WIS range: 0 (perfectly efficient) to 100 (critical waste, full isolation required).

THE 5-LEVEL INTERVENTION LADDER

WIS Range	Severity	System Action
0 to 20	Normal / Efficient	No action. Continue 15-minute monitoring cycle.
21 to 40	Low Waste	Soft nudge: WhatsApp/Telegram alert to class representative or lab assistant.
41 to 60	Medium Waste	Automated setpoint adjustment: AC raised 2 degrees C, Zone A lights dimmed. Admin alert sent.
61 to 80	High Waste	Relay trigger: Zone A lights off, AC to 28 degrees C, AV equipment enters sleep mode.
81 to 100	Critical Waste	Full DB isolation: All non-safety circuits cut. Event logged to audit trail.

Safety Veto Rule: If the ERP timetable shows a class is officially scheduled, the minimum WIS threshold to trigger any shutoff automatically rises by 20 points, preventing false positives on sparsely attended but legitimate sessions.

Adaptive Learning: GridSense re-calibrates all input weights every 30 days using logged outcome data. If manual override requests frequently follow automated interventions in a specific room, the WIS threshold for that room is raised by 10 points. The system learns the specific behavioral patterns of its campus over time.

THE 8 OPTIMIZATION MODULES (Ranked by Energy Impact)

MODULE A: Bio-Climatic Classroom Optimization and Daylight-First Scheduling Priority Rank: 1 Estimated Savings: 40% to 55% of total classroom energy consumption

A1: Daylight-First Room Allocation Each classroom is assigned a Daylight Autonomy Score based on its floor, window-to-wall ratio, and cardinal sun orientation. Morning sessions (8:30 AM to 12:30 PM) are prioritized into east and north-facing classrooms where natural daylight exceeds 300 lux, allowing ceiling lights to remain completely off. Afternoon sessions avoid unshaded west-facing rooms that absorb direct solar heat gain, reducing thermal load on AC compressors. Savings: 60% reduction in daytime artificial lighting usage through scheduling alone, with zero hardware.

A2: Occupancy-Scaled Air Conditioning Each student radiates approximately 100 Watts of body heat. A class of 60 generates 6 kW of internal heat; a class of 10 generates only 1.2 kW. The system reads live ERP headcount and cross-references outdoor weather API data every 15 minutes.

Setpoint Matrix:

- Full class (40 to 60 students) + Outdoor temp above 32 degrees C: Setpoint 24 degrees C, Fans at Speed 1 to 2
- Partial class (10 to 25 students) + Outdoor 24 to 30 degrees C: Setpoint 26 degrees C, Fans at Speed 3
- Mild weather / Rainy season below 24 degrees C: AC OFF, Fans only, full natural ventilation

Thermal Flywheel: At 15 minutes before class end, the AC compressor shuts off. The blower circulates residual cold air. The concrete thermal mass retains comfort until students leave. Savings: 30% to 45% reduction in HVAC power draw per classroom.

A3: Zoned Window-Row vs Interior-Row Lighting Ceiling lights are split into two independently controlled zones. Zone A (window-adjacent rows) locks off whenever outdoor ambient light exceeds 300 lux. Zone B (interior rows and blackboard strip) activates only when the room is occupied and

Zone A alone is insufficient. Savings: 50% reduction in classroom lighting load during daytime hours in perimeter classrooms.

A4: Smartboard and Audio-Visual Auto-Sleep If no touch interaction, HDMI signal, or wireless screen-casting is detected for 7 continuous minutes, the smartboard drops to under 3 Watts sleep mode. At the precise class-end timestamp, a network command (IP/RS232/CEC) forces complete shutdown of the smartboard, audio amplifier, and projector relay. Savings: 90% elimination of AV standby waste.

MODULE B: Spatial Building Consolidation and Wing Shutdown Priority Rank: 2 Estimated Savings: 35% to 50% of after-hours building infrastructure energy

The consolidation algorithm runs at 4:00 PM daily for after-hours planning. If Block A has 1 scheduled class and Block C already has 5, the lone class is relocated to Block C (subject to room capacity, equipment availability, and accessibility constraints). The entire Block A wing is then isolated at the main Distribution Board.

Corridor lights are split into 3 sequential relay zones. Only the zone immediately surrounding active classrooms is lit, eliminating 50 to 80 metres of empty hallway lighting.

Constraint-Aware Room Swapping checks:

- Class size vs room seating capacity
- Equipment requirements (smartboard, projector, specialized lab audio)
- Accessibility (students or faculty with mobility needs cannot be forced up staircases if elevators are offline)

Advance notice dispatched via WhatsApp bot or ERP push notification to class representative and professor with sufficient time to adjust.

MODULE C: Computer Laboratory Intelligent Power Management Priority Rank: 3 Estimated Savings: 28% to 40% of academic IT block energy

A lightweight background agent deployed on every lab PC (Windows or Ubuntu OS image) detects keyboard and mouse inactivity. After 15 minutes of no user input, the PC drops to S3 Sleep mode, drawing under 2 Watts instead of 80 to 120 Watts. At 6:00 PM, a central network broadcast script gracefully shuts down every unlocked, unattended machine across all labs.

Wake-on-LAN (WoL) allows the lab administrator to power all machines simultaneously the next morning with a single click. No smart plugs or additional hardware are required.

Turning off 60 PCs also eliminates 6 kW of internal heat generation, allowing the dedicated split AC to shut down completely.

MODULE D: Dynamic Elevator Staging with Mandatory 2-Lift Safety Floor Priority Rank: 4 Estimated Savings: 6,600 kWh per year per 4-lift bank (INR 59,400 annually)

The system never drops below 2 active lifts at any time, ensuring emergency evacuation capability, accessibility compliance, and mechanical redundancy.

Dispatch Matrix:

- Rush hours (8:30 to 9:30 AM, 12:30 to 1:30 PM, 4:30 to 5:30 PM): 4 lifts active, 0 in deep sleep
- Mid-day lecture hours (9:30 AM to 12:30 PM, 1:30 PM to 4:30 PM): 3 lifts active, 1 in deep sleep
- After-hours (5:30 PM to 8:30 AM and all weekends): 2 lifts active (mandatory minimum), 2 in deep sleep

Energy Calculations (Rounded Up):

- Running motor load: 8 kW per lift during transit
- Standby idle load: 0.6 kW per active parked lift
- Savings from deenergizing 2 lifts during 15 after-hours hours:
  - Standby saved: 2 lifts x 0.6 kW x 15 hours = 18 kWh per night
  - Unnecessary trip cycles saved: 20 trips x 0.025 hours x 8 kW = 4 kWh per night
  - Total conservative savings: 22 kWh per day
- Monthly savings: 22 x 30 = 660 kWh per month
- Annual savings (300 academic days): 22 x 300 = 6,600 kWh per year
- Financial savings at INR 9 per unit: INR 59,400 per year per elevator bank

MODULE E: Hostel Block Geyser Staggering and Solar-Aligned Water Pumping Priority Rank: 5 Estimated Savings: 25% to 35% of hostel peak demand costs and full avoidance of utility kVA penalty tariffs

All hostel wing geysers (2 to 3 kW each) fire simultaneously at 6:00 AM, pushing the campus above its contracted Maximum Demand (kVA) threshold and triggering penalty charges from the state electricity board (BESCOM, MSEDCL, TANGEDCO, TSSPDCL).

GridSense staggers geyser circuits wing-by-wing using programmable contactors:

- Wing 1: 5:30 to 6:30 AM
- Wing 2: 6:30 to 7:30 AM

Insulated water tanks retain heat for over 2 hours, so students in Wing 2 experience no reduction in hot water comfort.

Overhead water pump schedules are shifted from the morning grid-peak window to the campus solar generation peak (11:00 AM to 2:00 PM), directly offsetting grid draw with captive renewable generation.

MODULE F: Campus Water Cooler and Cold-Chain Compressor Interlocking Priority Rank: 6 Estimated Savings: 1,620 kWh per month (INR 14,580 per month)

Approximately 15 commercial water coolers (1.5 kW compressor each) across all 3 academic blocks run active refrigeration 24 hours per day, including through the night when buildings are completely locked.

Smart relay timers cut power to all cooler circuits at 7:00 PM and restore them at 7:00 AM on weekdays. Coolers are fully off on Sundays and declared holidays.

Insulated stainless cooling tanks retain chill for over 2 hours after shutdown, so late-staying students still receive cold water.

In the cafeteria, temperature probe sensors inside cold storage optimize compressor run cycles based on off-hours door seal integrity, preventing frost buildup and unnecessary short-cycling.

Savings calculation:

- 15 coolers x 1.5 kW x 12 hours off per day = 270 kWh per day baseline
- Conservative savings estimate: 54 kWh per day (rounded up)
- Monthly: 54 x 30 = 1,620 kWh per month
- Financial: 1,620 x INR 9 = INR 14,580 per month

MODULE G: Time-of-Day Heavy Infrastructure Load Shifting Priority Rank: 7 Estimated Savings: 20% to 30% reduction in peak-tariff electricity expenditure

Indian state electricity boards charge institutional consumers 20% to 40% higher tariff per unit during peak hours (6:00 PM to 10:00 PM).

Non-critical high-draw motors locked out between 6:00 PM and 10:00 PM:

- Sewage Treatment Plant aeration pumps
- Overhead water distribution pumps
- Central RO water purification plants

These loads are shifted to off-peak windows or to the solar-surplus hours (11:00 AM to 2:00 PM), directly offsetting grid draw with captive renewable generation and flattening the campus peak demand curve to prevent Maximum Demand penalty charges.

MODULE H: Enterprise WiFi Access Point Scheduled Power Reduction Priority Rank: 8 Estimated Savings: 1,800 to 2,700 kWh per month across a 150-AP campus

Each PoE access point draws 15 to 30 Watts continuously. Across 150 APs: 3 to 5 kW running 24 hours per day regardless of whether any student is on campus.

Between 11:00 PM and 6:00 AM in academic blocks, the central network switch controller (via PoE schedule) disables secondary 5 GHz radios and drops APs to low-power 2.4 GHz IoT-only beaconing mode (under 5 Watts per unit).

Hostel APs remain at full power. This saves 6 to 8 hours of full-load PoE draw per AP per night across all academic block access points.

SYSTEM ARCHITECTURE OVERVIEW

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RAW DATA INPUTS
  WiFi AP Client Count (SNMP)
  ERP Timetable + Attendance API
  Smart Sub-Meter (kW, kVAh per zone)
  Weather API (Temperature, Lux, Humidity)
  RFID Entry Logs
  Time-of-Day Tariff Calendar

GRIDSENSE SCORING ENGINE (every 15 minutes)
  Occupancy Sub-Score  (Weight 0.60)
  Climate Sub-Score    (Weight 0.23)
  Energy Sub-Score     (Weight 0.21)
  Schedule Sub-Score   (Weight 0.12)
  Final WIS: 0 to 100 per zone

DECISION ENGINE
  WIS 0 to 20   : Monitor only
  WIS 21 to 40  : Soft notification
  WIS 41 to 60  : Automated setpoint adjustment
  WIS 61 to 80  : Relay control trigger
  WIS 81 to 100 : Full DB isolation

OUTPUT LAYER
  LLM Executive Daily Digest (7:00 AM)
  Real-Time Campus Zone Dashboard
  WhatsApp / Telegram Notifications
  Monthly Savings Audit Report
```

TECHNOLOGY STACK

Layer	Technology	Already on Campus
Occupancy Detection	Enterprise WiFi SNMP + ERP API	Yes (99% of campuses)
Power Monitoring	Smart sub-meters at Distribution Boards	Partially
Climate Data	Weather API (OpenMeteo / IMD free tier)	Cloud API, zero cost
Scoring Engine	Python FastAPI backend, Supabase PostgreSQL	Software only
Control Layer	Smart contactors at DB panels	New (moderate one-time cost)
Notifications	WhatsApp / Telegram bot API	Cloud API, zero cost
Reporting	LLM API (Gemini free tier)	Cloud API, zero cost
Dashboard	React frontend on Render.com	Software only

SECTION 5: IMPACT AND FEASIBILITY

Combined Monthly Savings Table

Module	Monthly kWh Saved	Monthly INR Saved	New Hardware
A: Classroom Bio-Climatic	3,800 kWh	INR 34,200	None
B: Building Consolidation	3,200 kWh	INR 28,800	DB contactors per wing
C: Computer Lab Management	2,400 kWh	INR 21,600	None (software only)
D: Elevator Staging	660 kWh	INR 5,940	Contactor relay on lift DB
E: Hostel Geyser + Pumps	1,800 kWh	INR 16,200 + penalty savings	Programmable contactors
F: Water Cooler Interlocking	1,620 kWh	INR 14,580	Smart relay timers
G: ToD Load Shifting	820 kWh	INR 7,380 + tariff savings	Contactor interlocks
H: WiFi AP Scheduling	700 kWh	INR 6,300	None (PoE switch schedule)
TOTAL	**15,000 kWh/month**	**INR 1,35,000/month**	

Key Impact Numbers

- Annual cost savings: INR 15 Lakhs or more (conservative estimate)
- Carbon reduction: 48 Tons of CO2 equivalent prevented annually (at CEA 2023 Indian grid factor of 0.82 kg CO2 per kWh)
- Hardware ROI period: 4 to 6 months (funded entirely from electricity bill savings)
- Phase 1 deployment timeline: 2 weeks (software only, zero new hardware)
- Full deployment timeline: 6 weeks

Feasibility: What Already Exists on Campus

- Enterprise WiFi controller with SNMP API (Aruba, Cisco, UniFi)
- ERP system with timetable and attendance data
- Main smart meter at the campus incomer panel
- Existing network infrastructure for LAN communication
- Academic calendar and holiday data

Minimum Viable Data Principle

GridSense delivers actionable value from Day 1 using only the WiFi controller API and ERP timetable data, both of which every modern Indian engineering college already operates. The system incrementally incorporates more data sources as Phase 2 hardware is commissioned, improving WIS accuracy with each addition.

Scalability

GridSense is not built for one campus. The scoring engine, weight system, and intervention ladder are fully parameterizable. Any campus with a WiFi controller API and ERP system can be onboarded within 2 weeks. Weights are tuned per campus based on building layout, tariff structure, and occupancy patterns. The adaptive learning engine ensures GridSense becomes more accurate with every 30-day recalibration cycle.

PROMPT FOR PDF DOCUMENT (Claude / ChatGPT)

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You are a professional technical document designer. Create a comprehensive, well-structured PDF-ready document based on the following knowledge base. The document is a solution deck for a hackathon.

PROJECT DETAILS:
- Project Name: GridSense
- Tagline: Campus Intelligent Energy Optimization Platform
- Team: Quantum Minds
- Institution: Jain Institute of Technology (JIT)
- Challenge: No. 12 / The Campus That Wastes Energy After Hours
- Event: Hackathon O1

DOCUMENT STRUCTURE:
The document must contain exactly these 5 sections in this order:

1. EXPLAIN THE CHALLENGE
2. WHAT IS THE PROBLEM
3. GAPS AND CHALLENGES
4. SOLUTION AND ARCHITECTURE
5. IMPACT AND FEASIBILITY

DESIGN REQUIREMENTS:
- Professional dark-theme design with a clean cover page showing the team name, institution, challenge number, and project name GridSense
- Each section must start on a new page
- Use clear headings, subheadings, tables, and bullet points
- Include the Waste Index Score formula and the 5-level intervention ladder as a visual table
- Include the 8 optimization modules (A through H) with their priority ranks and savings
- Include the combined savings table in Section 5
- No emojis anywhere in the document
- No casual language. Keep the tone technical and professional throughout

KNOWLEDGE BASE:
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PROMPT FOR POWERPOINT PRESENTATION (Claude / ChatGPT /

Gamma.app)

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You are a professional presentation designer. Create a detailed PowerPoint presentation
script and slide-by-slide breakdown based on the following knowledge base. This is a
hackathon solution deck.

PROJECT DETAILS:
- Project Name: GridSense
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SLIDE STRUCTURE (create exactly these slides):

SLIDE 1: COVER
- Title: GridSense
- Subtitle: Campus Intelligent Energy Optimization Platform
- Team: Quantum Minds
- Institution: Jain Institute of Technology
- Challenge: No. 12 / The Campus That Wastes Energy After Hours

SLIDE 2: EXPLAIN THE CHALLENGE
- The situation on campus at 10:00 PM
- 3 key statistics: 40% after-hours waste, INR 8L+ monthly bill, 30-45% preventable
- The fundamental mismatch: static energy systems vs dynamic human presence

SLIDE 3: WHAT IS THE PROBLEM
- 4 zones of waste: Academic Blocks, Computer Labs, Hostels, Infrastructure
- Each with its root cause
- The phantom load phenomenon: empty rooms at full power

SLIDE 4: GAPS AND CHALLENGES
- Gap 1: The Raw Data Trap (bulk bills vs actionable insight)
- Gap 2: The Hardware Cost Barrier (full IoT retrofit too expensive)
- Gap 3: The All-or-Nothing Dilemma (binary policies fail)
- Gap 4: Spatial Fragmentation is Invisible to current BMS systems

SLIDE 5: INTRODUCING GRIDSENSE
- What is GridSense: a weighted multi-input scoring engine
- Core principle: Waste Index Score (WIS) of 0 to 100 per zone per 15 minutes
- The higher the score, the more aggressive the intervention

SLIDE 6: THE SCORING ENGINE
- 4 input categories with their weights:
  Occupancy: 0.60
  Climate: 0.23
  Energy Draw: 0.21
  Schedule: 0.12
- Key inputs in each category

SLIDE 7: THE 5-LEVEL INTERVENTION LADDER
- Table showing WIS ranges 0-20, 21-40, 41-60, 61-80, 81-100
- Each level's severity and automated action
- Safety Veto Rule: ERP-scheduled classes raise threshold by 20 points

SLIDE 8: THE 8 OPTIMIZATION MODULES (Overview)
- A through H listed with one-line description each and priority rank

SLIDES 9 to 12: MODULE DEEP DIVES (group them 2 per slide)
- A + B
- C + D
- E + F
- G + H
Each with savings estimate and key mechanism

SLIDE 13: SYSTEM ARCHITECTURE DIAGRAM
- Data inputs flow into GridSense Scoring Engine
- Engine outputs WIS per zone
- Decision Engine dispatches 5-level interventions
- Output Layer: Dashboard, notifications, LLM report

SLIDE 14: IMPACT AND FEASIBILITY
- Total: 15,000 kWh saved per month, INR 1,35,000 per month, INR 15L+ per year
- 48 tons CO2 reduced annually
- Hardware ROI: 4 to 6 months
- Phase 1 deployment: 2 weeks, software only, zero new hardware

SLIDE 15: FEASIBILITY AND SCALABILITY
- What already exists on campus (WiFi SNMP, ERP, smart meter)
- Minimum Viable Data principle
- Scalable to any campus with WiFi controller and ERP within 2 weeks

SLIDE 16: CLOSING SLIDE
- GridSense
- Quantum Minds | Jain Institute of Technology
- Challenge 12 | Hackathon O1
- Tagline: Track. Understand. Eliminate. Campus energy waste ends here.

DESIGN REQUIREMENTS:
- Dark professional theme
- Each slide should have a maximum of 5 to 6 bullet points
- All numbers and statistics must be prominently displayed
- No emojis
- Tables where relevant (intervention ladder, savings table)
- Keep tone technical and confident

KNOWLEDGE BASE:
[PASTE THE ENTIRE KNOWLEDGE BASE TEXT HERE BELOW THIS LINE]
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GridSense Knowledge Base v1.0 | Quantum Minds | Jain Institute of Technology | Challenge 12